Novelty Statement: Recent co-author work cited in this article (Wion et al. 2025) is the seed cone dataset used in the analysis of bird abundance. This is the first time this dataset has been combined with data on wildlife abundance linked to this key food resource.

We provide a PDF of this article for reviewers.

School of Informatics, Computing, and Cyber Systems

Ana Miller-ter Kuile
ana.miller-ter-kuile@nau.edu

School of Informatics
1295 Knoles Drive
Flagstaff, AZ 86011

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Dear Editorial Board,

Populations of species across the globe are declining due to a suite of forces, including habitat and climate change. However, the decline of interaction partners may be an under-examined driver of population trends. Examining the effects of species interactions on populations is challenging given the need for broad-scale and long-term datasets on multiple species. We attach a manuscript entitled "**Caching in: A seed dispersal mutualism signals the decline of a songbird-pine system**" in which we address this knowledge gap by quantifying relationships between seed abundance and bird abundance in a system where both interacting partners are declining – pinyon jays and two-needle pinyon pine in the southwestern USA. We overcame data limitations by leveraging complementary data on both interacting partners: over 800,000 observations from the eBird citizen science program and a regionally dense dataset on seed availability that spans the past few decades. We feel this work is suitable for *Ecology Letters* because of the pressing challenge of mitigating global biodiversity loss and the need for approaches that strengthen our predictive understanding of ongoing population declines. In this specific system, our study is timely given that pinyon jays are being considered for listing under the Endangered Species Act. Should the species be listed, listing will transform conservation priorities across most of the western USA with impacts on many species and ecosystems.

This study used a Bayesian statistical modeling approach never used before in examining plant-animal interactions (N-mixture model combined with a stochastic antecedent modeling framework) to examine the temporal legacies and potential anticipatory responses of species interactions. We found that pinyon jay abundance is directly linked to seed abundance ("immediate" response) and that this effect is comparable to or of greater importance than many climate and habitat drivers. Further, the effect of cone availability has a three-year legacy on abundance of pinyon jays ("delayed" response). While we focus on one ecological system of conservation concern, the modeling approach we developed using broad-scale citizen science data has broad applicability across systems with interacting partners. In particular, in many systems, the temporal legacies of species interactions may be key to determining future population, community, and ecosystem states. We develop novel data preparation (e.g., spatial uncertainty in eBird citizen science data) and statistical (e.g., N-mixture + stochastic antecedent modeling) methods that have broad applicability to long-term ecological datasets.

Novelty Statement: Recent co-author work cited in this article (Wion et al. 2025) is the seed cone dataset used in the analysis of bird abundance. This is the first time this dataset has been combined with data on wildlife abundance linked to this key food resource.

This manuscript has not been previously published elsewhere. Thank you for considering our work.

Sincerely,

Ana Miller-ter Kuile, Ph.D.

Title: Caching in: A seed dispersal mutualism signals the decline of a songbird-pine system

Author List: Ana Miller-ter Kuile[1,2], Andreas P. Wion[3], Kyle C. Rodman[4], Jamie S. Sanderlin[2], Kiona Ogle[1]

Author Affiliations:

[1] School of Informatics, Computing, and Cyber Systems, Northern Arizona University, Flagstaff, Arizona, USA

[2] USDA Forest Service, Rocky Mountain Research Station, Flagstaff, Arizona, USA

[3] Forest Stewards Guild, Santa Fe, New Mexico, USA

[4] Ecological Restoration Institute, Northern Arizona University, Flagstaff, Arizona, USA

Author Emails:

Ana Miller-ter Kuile: ana.miller-ter-kuile@nau.edu

Andreas P. Wion: apwion.work@gmail.com

Kyle C. Rodman: kyle.rodman@nau.edu

Jamie S. Sanderlin: jamie.l.sanderlin@usda.gov

Kiona Ogle: kiona.ogle@nau.edu

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Corresponding author: Ana Miller-ter Kuile, ana.miller-ter-kuile@nau.edu, Northern Arizona University School of Informatics, Computing, and Cyber Systems, 1295 Knoles Drive, Flagstaff, Arizona 86011

Statement of Authorship: AM-tK, APW, and KCR conceived the ideas; AM-tK, KO, and JSS developed statistical methods; AM-tK and KCR developed novel data layers and ran statistical models; AM-tK led the writing of the manuscript with support from APW, KCR, and KO. All authors contributed to the drafts and gave final approval for publication.

Data Accessibility Statement: Data and code for this paper are available on Zenodo (Miller-ter Kuile *et al.* 2026).